Hints: Linear algebra – 9. Matrices over $\mathbb{Z}$ and over a finite field

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. An invertible matrix is a list of columns, each outside the span of the previous ones: there are $q^n - q^k$ choices for column $k+1$. For subspaces, count ordered k-tuples of independent vectors in two ways: directly, and as (number of subspaces)$\times|\mathrm{GL}_k(\mathbb{F}_q)|$. This gives $\prod_{i<k} \frac{q^n-q^i}{q^k-q^i}$.

Hint 2. $X = |\ker(g-E)|$, and X^k is the number of k-tuples $(v_1, \dots, v_k)$ fixed by g. By Burnside's lemma $\mathsf{E}X^k$ is the number of orbits of $\mathrm{GL}_n(\mathbb{F}_q)$ on $(\mathbb{F}_q^n)^k$.

Hint 3. A k-tuple is a linear map $\mathbb{F}_q^k \to \mathbb{F}_q^n$. For $n \geqslant k$ two tuples lie in one orbit if and only if they have the same kernel (extend a basis of the image to a basis of $\mathbb{F}_q^n$). So the orbits correspond to the subspaces of $\mathbb{F}_q^k$, and $\mathsf{E}X^k = \sum_{j=0}^k \binom{k}{j}_q$. For $k=1$: $\mathsf{E}X = 2$; for $k=2$: $\mathsf{E}X^2 = 1 + (q+1) + 1 = q+3$, so $\operatorname{Var} X = q-1$.

Problem 2

Hint 1. The polynomial $x^n + 1$ has distinct roots and annihilates A. So A is diagonalisable over $\mathbb{C}$, and its eigenvalues are roots of unity with $\lambda^n = -1$. The characteristic polynomial is a rational quadratic.

Hint 2. If the eigenvalues are rational, both equal -1 and $A = -E$. Otherwise the characteristic polynomial is a cyclotomic polynomial of degree 2: Φ_3, Φ_4 or Φ_6. Rule out Φ_3 (no cube root of unity has a power equal to -1). By Cayley–Hamilton, Φ_4 gives $A^2 = -E$ and Φ_6 gives $A^3 = -E$.

Problem 3

Hint 1. For $X \in \mathrm{SL}_2(\mathbb{Z})$, Cayley–Hamilton gives $X^2 = (\operatorname{tr} X)X - E$. The traces drive everything.

(a)

Hint 1. Try $\operatorname{tr} A = -1$, $\operatorname{tr} B = 1$, $\operatorname{tr} C = 0$. Then $A^2 + B^2 = B - A - 2E$ and $C^2 = -E$, so you need $B = A + E$. For example $A = \begin{pmatrix} -1 & -1 \\ 1 & 0 \end{pmatrix}$, $B = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}$, $C = \begin{pmatrix} -1 & -2 \\ 1 & 1 \end{pmatrix}$.

(b)

Hint 1. Take traces: $\operatorname{tr} X^4 = t^4 - 4t^2 + 2$ for $t = \operatorname{tr} X$. This gives $a^4 + b^4 - c^4 - 4(a^2 + b^2 - c^2) + 2 = 0$ for the traces a, b, c.

Hint 2. Modulo 4, fourth powers are 0 or 1, so a, b are odd and c is even. Then work modulo 16: $a^2 + b^2 \equiv 2 \pmod 8$ and $c^2 \equiv 0$ or $4 \pmod 8$. The left side is $\equiv 4 - 8 \not\equiv 0 \pmod{16}$.

Problem 4

Hint 1. The matrix is $J + D$ with $D = \operatorname{diag}(k^2-1)_{k=2}^N$, $N = p + 7$. So $\det = \prod(k^2-1) \cdot (1 + \sum 1/(k^2-1))$. Telescope both.

Hint 2. You should get $\det = ((N-1)!)^2 \, \dfrac{7N^2+3N-2}{8}$. For $p \geqslant 7$, p divides $(p+6)!$ exactly once. So you need $p \mid 7 \cdot 7^2 + 3 \cdot 7 - 2 = 362 = 2 \cdot 181$. Check $p = 2, 3, 5$ directly.

Hint 3. Answer: $p \in \{2, 3, 5, 181\}$.

Problem 5

Hint 1. The entries $a_{\sigma(i)i}$ lie in distinct rows and columns. Hence $f(\varepsilon) = \det$ of the modified matrix is a polynomial in $\varepsilon(1), \ldots, \varepsilon(n)$ of degree at most 1 in each variable. What is the coefficient of $\varepsilon(1) \cdots \varepsilon(n)$?

Hint 2. That coefficient is $\operatorname{sgn} \sigma = \pm 1$. For a multilinear polynomial, $\sum_{\varepsilon \in \{0,1\}^n} (-1)^{n-|\varepsilon|} f(\varepsilon)$ equals this top coefficient. If m divided every $f(\varepsilon)$, it would divide ± 1.

Problem 6

Hint 1. For $p = 2$ compare orders. For odd p, look at $u = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ of order p. In S_p an element of order p is a p-cycle. What is its centraliser?

Hint 2. The centraliser of a p-cycle in S_p has order p. The centraliser of u in $\mathrm{GL}_2(\mathbb{Z}/p)$ contains all $\begin{pmatrix} a & b \\ 0 & a \end{pmatrix}$, which is $p(p-1)$ elements. An injective homomorphism maps $C(u)$ injectively into $C(\varphi(u))$.

Problem 7

Hint 1. Take logarithms of the (positive) entries. The operations become: add the other row (column) to a row (column), or subtract it.

Hint 2. These operations preserve the determinant of the matrix of logarithms. It is $\ln 2(\ln 4 - \ln 3)$ at the start and its negative at the target, and it is non-zero because $\ln 4 \neq \ln 3$. Answer: no.

Problem 8

Hint 1. Rank-one reduction (Wedderburn): $\operatorname{rank}(A - a e_i e_j^T) < \operatorname{rank} A$ if and only if $e_i \in \operatorname{Im} A$, e_j^T is in the row space and $a\, e_j^T G e_i = 1$, where G is any generalised inverse ($AGA = A$). For invertible A this says $(A^{-1})_{ji} = 1/a_{ij}$.

Hint 2. Read off the (i,i) entry of the idempotent AG. Since $e_i = Ax_0$, it equals 1; on the other hand it is $\sum_j a_{ij} G_{ji}$, a sum of ones over the non-zero entries of row i. So the number of non-zero entries of each non-zero row is $\equiv 1 \pmod p$. For $p > 2014$ this forces exactly one.

Hint 3. For $p \leqslant 2013$ a counterexample: $J - E$ of size $m = p + 2$ has inverse $\frac{1}{m-1} J - E = J - E$ over $\mathbb{F}_p$. Put it in a block-diagonal matrix with an identity block. Answer: exactly the primes $p \geqslant 2017$.

Problem 9

Hint 1. For $j \geqslant 1$, $i^j \bmod 3$ depends only on $i \bmod 3$ and the parity of j. Show that rows i and $i + 6$ coincide, so $\det A = 0$ for $n \geqslant 7$.

Hint 2. Compute the small cases by hand (only residues $0, 1, 2$ appear) and find a dependency among the rows for $n = 6$. Answer: $n = 5$ (indeed $\det A = 12$ there).

Problem 10

Hint 1. Orient the edges and let D be the $n \times m$ incidence matrix, so that $L = DD^T$. Delete a row of D to get D_0 and apply Cauchy–Binet to $\det(D_0 D_0^T)$.

Hint 2. An $(n-1) \times (n-1)$ minor of D_0 is ± 1 if the chosen edges form a spanning tree, and 0 otherwise. A cycle gives dependent columns; for a tree, expand at a leaf.

Hint 3. For $L(K_n) = nE - J$: the entries generate 1, so $d_1 = 1$. Modulo n, $L \equiv -J$ has rank 1, so all $k \times k$ minors are divisible by n^{k-1}. Moreover $d_1 \cdots d_{n-1} = n^{n-2}$ (Cayley). Hence the Smith form is $\mathrm{diag}(1, n, \ldots, n, 0)$ and the critical group is $(\mathbb{Z}/n)^{n-2}$.